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Experiment 4

Momentum and Impulse

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Worksheet

I. DISCUSSION

Glider Mass (g)	Photogate Flag Width (cm)
202.30 ± 0.05	3.80 ± 0.05

Table 1.1. Glider and Flag measurements. The setup of the experiment uses a glider with a photogate flag attached on top that passes through a photogate to impact a force sensor.

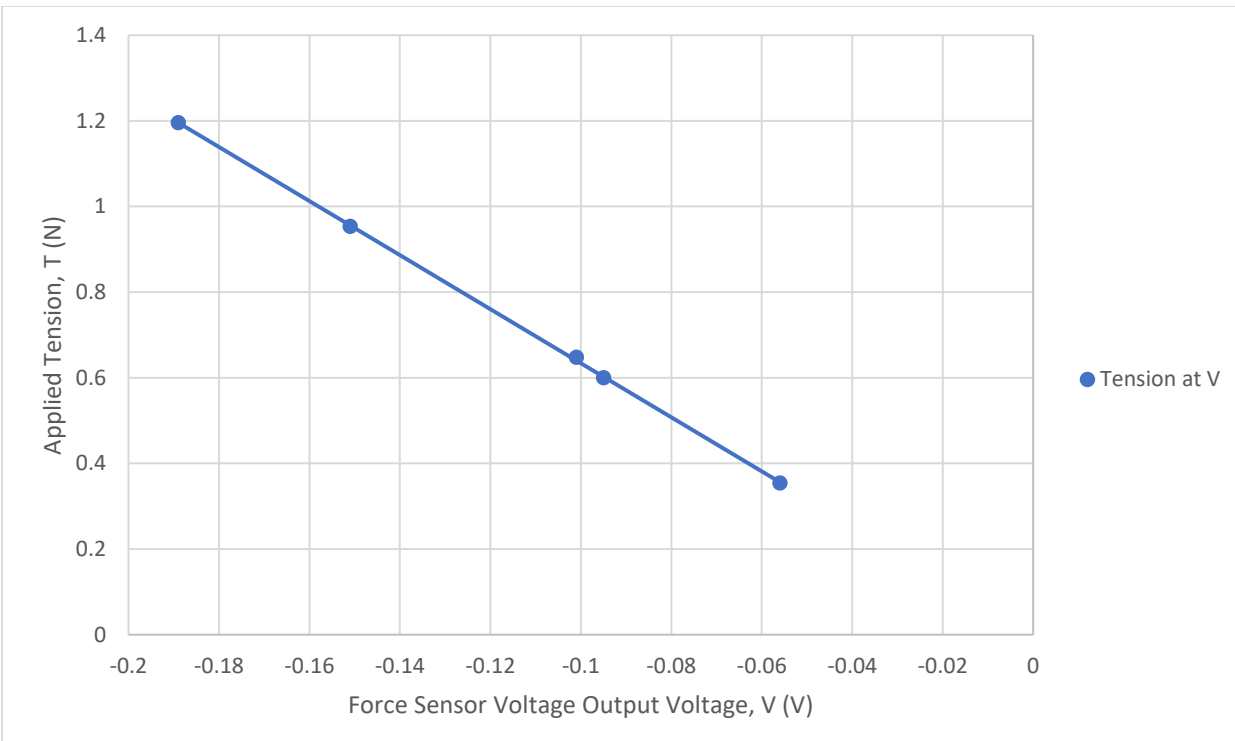

Figure 1.1. Force Sensor Calibration. Graph of the calculated tension experienced by the force sensor due to measured voltage values based on different mass values with the following best-fit line: $T = (-6.3119 \pm 0.0481)V + (0.0028 \pm 0.0061)$. The calibration constant is the slope of the best-fit line: $(-6.3119 \pm 0.0481)(N/V)$.

Trial Speed	Initial Velocity (m/s)	Final Velocity (m/s)
Fast	– 0.38 ± 0.05	0.22 ± 0.05
Slow	– 0.15 ± 0.05	0.08 ± 0.05

Table 1.2. Initial and final velocities of the glider based on speed. Velocities of the glider were recorded using a photogate with a negative initial velocity and a positive final velocity due to the change in direction of motion and the negative voltage output of the force sensor after the glider collides with the force sensor.

Calculation of Impulse

Method 1 – Direct Computation of Momentum

Impulse (J) is the change of momentum where momentum (p) is given by the formula: $p = mv$. Therefore, impulse can be derived as the following equation:

$$J = \Delta p = m\Delta v = m(v_f - v_i)$$

Additionally, the uncertainty of (J) can be calculated using the following equation:

$$\delta J = |J_{best}| \sqrt{\left(\frac{\delta m}{|m_{best}|}\right)^2 + \left(\frac{\sqrt{(\delta v_f)^2 + (\delta v_i)^2}}{|v_f - v_i|_{best}}\right)^2}$$

Trial Speed	Impulse (N*s)
Fast	0.1214 ± 0.0143
Slow	0.0465 ± 0.0143

Table 1.3. Impulse calculations using direct computation of momentum.

Method 2 – Riemann Sum Integration

Impulse (J) can also be calculated by integrating a plot of momentum over a period of time in which the momentum changes via Riemann Sum. With this method, impulse is equivalent to the sum of the force in a time interval $(F(t))$ multiplied by the time interval (Δt):

$$\Delta P = \int_{t_1}^{t_n} dt F(t) \approx \Delta t \sum_{i=1}^{n} F(t_i)$$

$$\Delta t = t_{i+1} - t_i$$

Trial Speed	Impulse (N*s)
Fast	0.1370 ± 0.0481
Slow	0.0467 ± 0.0481

Table 1.4. Impulse calculations using Riemann Sum Integration. $\boldsymbol{F(t_i)}$ was calculated by multiplying the calibration constant in Figure 1.1 and the measured voltages of the force sensor. Assuming the fractional uncertainty of the impulse values and the calibration constant value are the same, the calibration constant uncertainty serves as the impulse uncertainty.

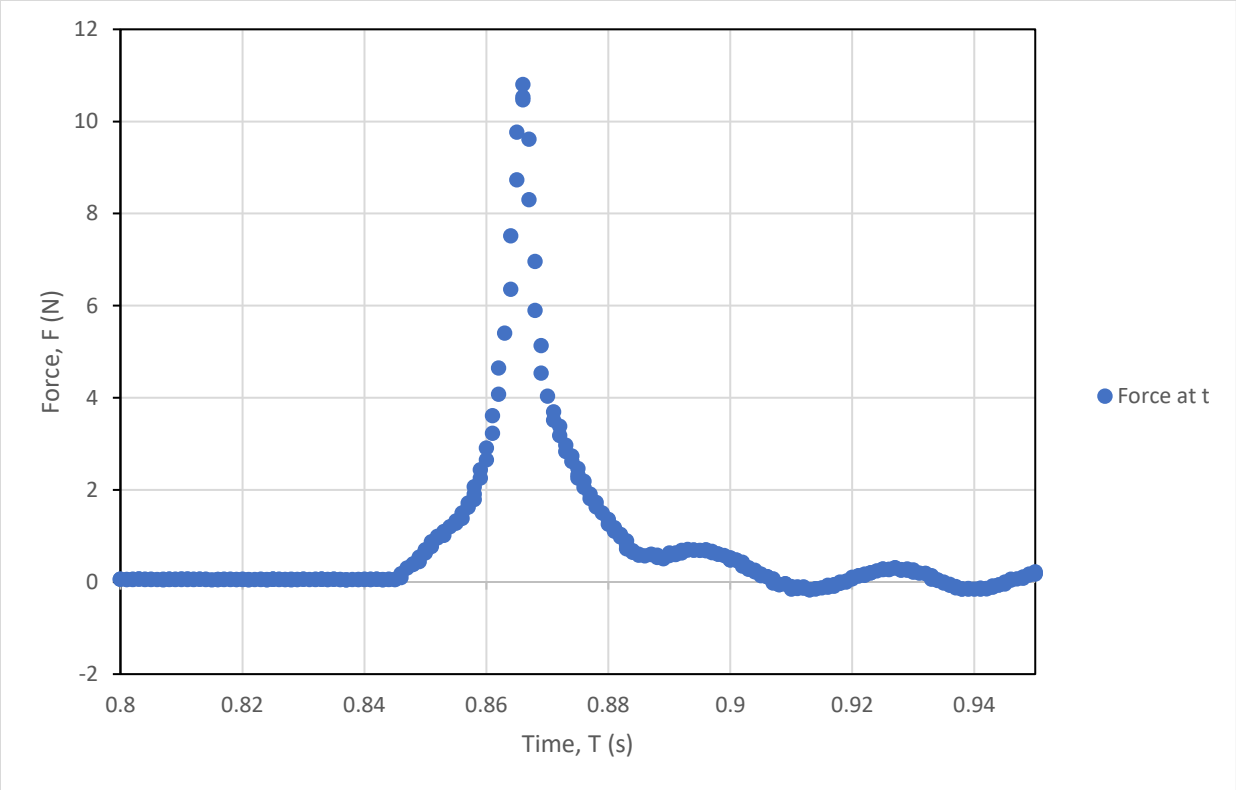

Figure 1.2. Graph of the force at time t experienced by the glider during the fast collision.

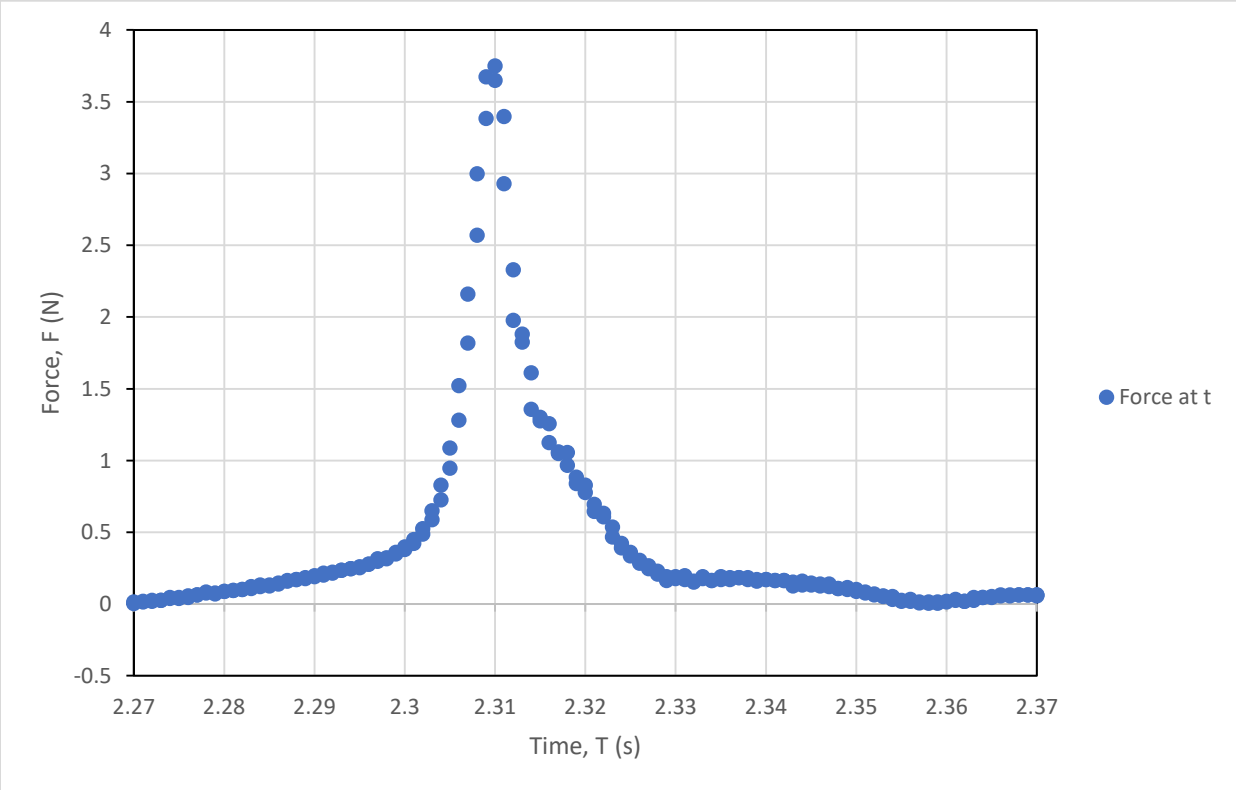

Figure 1.3. Graph of the force at time t experienced by the glider during the slow collision.

Results

Method	Trial Speed	Impulse (N*s)
1	Fast	0.1214 ± 0.0014
	Slow	0.0465 ± 0.0014
2	Fast	0.1370 ± 0.0481
	Slow	0.0467 ± 0.0481

Table 1.5. Impulse calculations for both trials using different methods.

Table 1.5 shows the results for the fast and slow collisions. For a given object in motion, higher differences in velocities at two points in time should produce higher impulses than lower differences. The data in Table 1.5 confirms these two expectations due to the higher impulses during the fast collisions than the slow collisions for both methods. The glider had an impulse difference of 0.0749 for method 1 and 0.0903 for method 2. Additionally, both fast and slow collisions produced impulse values that corresponded within the margins of error for each trial speed. The glider had an impulse difference of 0.0156 for the fast collisions and 0.0002 for the slow collisions. Notably, the faster collisions produced a more significant “ringing” effect within the force sensor than the slow collisions. This effect is shown in the aforementioned higher impulse difference for the fast collisions than the slow collisions and in Figure 1.2.

II. EXTRA CREDIT

Trial Speed	Initial Velocity	Final Velocity
Fast – Glider 1	– 0.533386 ± 0.000005	0.082730 ± 0.000005
Fast – Glider 2	– 0.535174 ± 0.000005	0.147905 ± 0.000005
Slow – Glider 1	– 0.296333 ± 0.000005	0.029789 ± 0.000005
Slow – Glider 2	– 0.347521 ± 0.000005	0.038442 ± 0.000005

Table 1.6. Measured initial and final velocities of two gliders with different speeds. The table shows velocity data on the collision of two gliders. The fast trial used two gliders with bumpers while the slow trial did not use any bumpers.

Trial Speed	C_R
Fast – Glider 1	0.155103 ± 0.000009
Fast – Glider 2	0.276368 ± 0.000010
Slow – Glider 1	0.100525 ± 0.000017
Slow – Glider 2	0.110618 ± 0.000014

Table 1.7. Calculated Values of $\boldsymbol{C_R}$ based on the trial speed of each collision. The coefficient of restitution (C_R) is defined as the ratio of the final relative speed to the ratio of the initial relative speed of the colliding objects. Therefore, $C_R = V_f/V_i$.

Table 1.7 shows that the $\boldsymbol{C_R}$ for the fast trial speeds are higher than the slow speeds by an average value of 0.110164. These values suggest that the faster collision with bumpers attached to the gliders are more elastic and also conserved more energy and momentum than the slower collision with no bumpers. Additionally, the large difference in $\boldsymbol{C_R}$ by a margin 0.121265 for the fast collision suggests that one of the two bumpers conserves energy better than the other.

Trial Speed	Force (N)	Impulse (N*s)
Fast – Glider 1	0.038396 ± 0.000393	0.124640 ± 0.000031
Fast – Glider 2	0.047196 ± 0.000393	0.138187 ± 0.000034
Slow – Glider 1	0.010757 ± 0.000082	0.065975 ± 0.000016
Slow – Glider 2	0.015068 ± 0.000082	0.078080 ± 0.000019

Table 1.8. Calculated values force and impulse. The change in kinetic energy was treated at force and the change in momentum was treated as impulse experienced by the two gliders. Uncertainty was calculated using data regression for force and the impulse uncertainty equation.

By taking the difference of the force and impulse values for each trial, the conservation of energy and momentum are quantified. For the fast collision, energy is conserved at 0.008800 N and momentum is conserved at 0.013547 N*s. For the slow collision, energy is conserved at 0.004311 N and momentum is conserved at 0.012105 N*s. As expected, the faster and elastic collisions conserve energy and momentum better than the slower and inelastic collision.

Mini-Report

Introduction

In 1687, Isaac Newton discussed the concept of force, momentum, and impulse in his work, the Mathematical Principles of Natural Philosophy. Newton defined force as the rate of change of momentum. With this definition, impulse is derived as the integral of force over time as well as the integral of the change of momentum over an interval of time. The objective of the experiment is to quantify impulse as a vector quantity by examining the collision of a glider system with different velocities. In addition, the experiment uses two methods to compare and verify the calculated quantities of impulse. The first method tracks the velocity of the glider during pre- and post-collision using a photogate. With these two quantities, impulse is directly calculated using the measured mass of the glider and the change in velocity at the two points of time. The second method uses a data acquisition tool CAPSTONE to record the instantaneous force experienced by the glider during pre- and post-collision. By calculating the average force for each time interval, impulse is calculated using Riemann Sum Integration where the average force values are summed and multiplied with the measured time interval. By using these two methods for two different glider velocities, impulse is quantified and verified.

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Methods

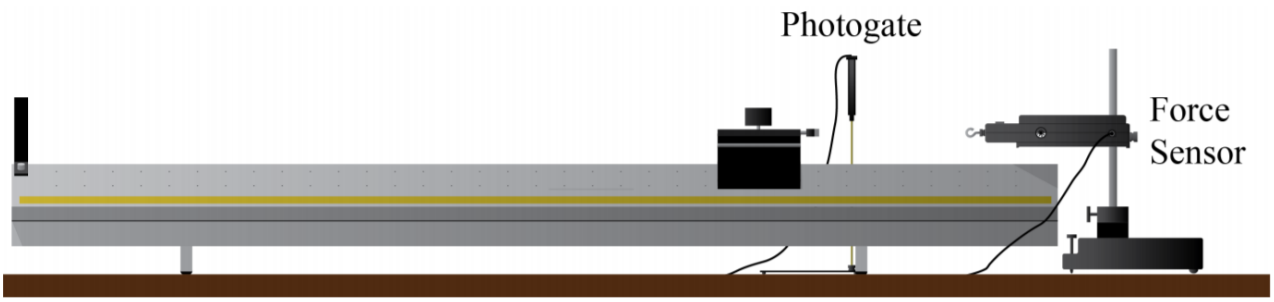

Figure 1.6. Experiment Setup for data acquisition. The experiment setup of the experiment will comprise of a glider mass, a photogate flag, a photogate, an impact force sensor.

The initial portion of the experiment is the calibration of lab equipment to ensure that the proper data is recorded while minimizing uncertainty. The sources of uncertainty in this experiment will primarily originate in the air track and the impact force sensor. To minimize the uncertainty due to the air track, the glider mass was placed on the air track and the gradient of the track was adjusted to minimize the movement of the glider. To minimize the uncertainty due to the impact force sensor, the force sensor must be calibrated using five different masses. With

Capstone, the voltage values produced by the force sensor for each of the five chosen masses were recorded. The data will produce a linear plot with the slope as the calibration constant, which will be used for converting recorded voltage values to force values. In addition to calibration, the mass value of the glider was measured for momentum calculations and the length of the photogate flag for the PASCO timer setup.

The second portion of the experiment involves the DAQ setup for Capstone. Using a PASCO DAQ system, Capstone was configured to record time, voltage values of the impact sensor, and glider speed tracked by the photogate. To ensure that the recorded data is precise enough for analysis, Capstone was configured to measure data values with enough precision digits. Additionally, Capstone was configured to record data at a sampling rate of equal to or more than 2 kHz to ensure that enough is data is recorded to produce a force over time plot for each trial. Additionally, the sample rate must not exceed 10 kHz to due to issues with Capstone's ability to record data samples exceeding the aforementioned frequency.

The third portion of the experiment involves the glider collision trials based on glider speed. Two trials were conducted to ensure that the data for the impulse calculations were based on two different velocities. Both trials were conducted in the same manner except the glider bumpers were different for each trial. By doing so, the measured data can be categorized under a slow velocity and a fast velocity for comparison. With the two data sets, the voltage values are converted to force values using the calibration constant calculated during the force sensor calibration. To reduce the uncertainty in the force data sets, the average force values for two time points were calculated.

The last portion of the experiment is the analysis of the data sets using two methods. The first method directly computes the impulse of the glider for both slow and fast collisions using the following formula for the rate of change of momentum: $J = \Delta p = m(v_f - v_i)$. The second method integrates the area under the peak of a force over time graph of the slow and fast collision data sets via Riemann Sum Integration. This method computes the impulse by multiplying the sum of the average force over the peak time period and the difference of each consecutive point in time for each average force: $J = \Delta P = \int_{t_1}^{t_n} dt F(t) \approx \Delta t \sum_{i=1}^{n} F(t_i)$. The graphs for the second method was plotted in Figures 1.2 and 1.3. The resulting impulse values for method 1 should correspond to the impulse values for method 2. For uncertainty, the following formula was used for method 1: $\delta J = |J_{best}| \sqrt{\left(\frac{\delta m}{|m_{best}|}\right)^2 + \left(\frac{\sqrt{(\delta v_f)^2 + (\delta v_i)^2}}{|v_f - v_i|_{best}}\right)^2}$. For method 2, the fractional uncertainty of the calibration constant values and the calculated impulse values were assumed to be the same. Therefore, the calibration constant uncertainty was used as the impulse uncertainty.

Bibliography

1. Campbell, W. C. et al. Physics 4AL: Mechanics Lab Manual (ver. June 27, 2018). (Univ. California Los Angeles, Los Angeles, California).